



Building & Scaling Custom HEOR Modelling Software within Industry

From spreadsheets to an R ecosystem in an enterprise environment



Robert Smith, Yevgeniy Samyshkin
R for HTA – Exeter - Wednesday 24th June 2026 (Remote)



Yevgeniy Samyshkin is an employee and shareholder of GSK

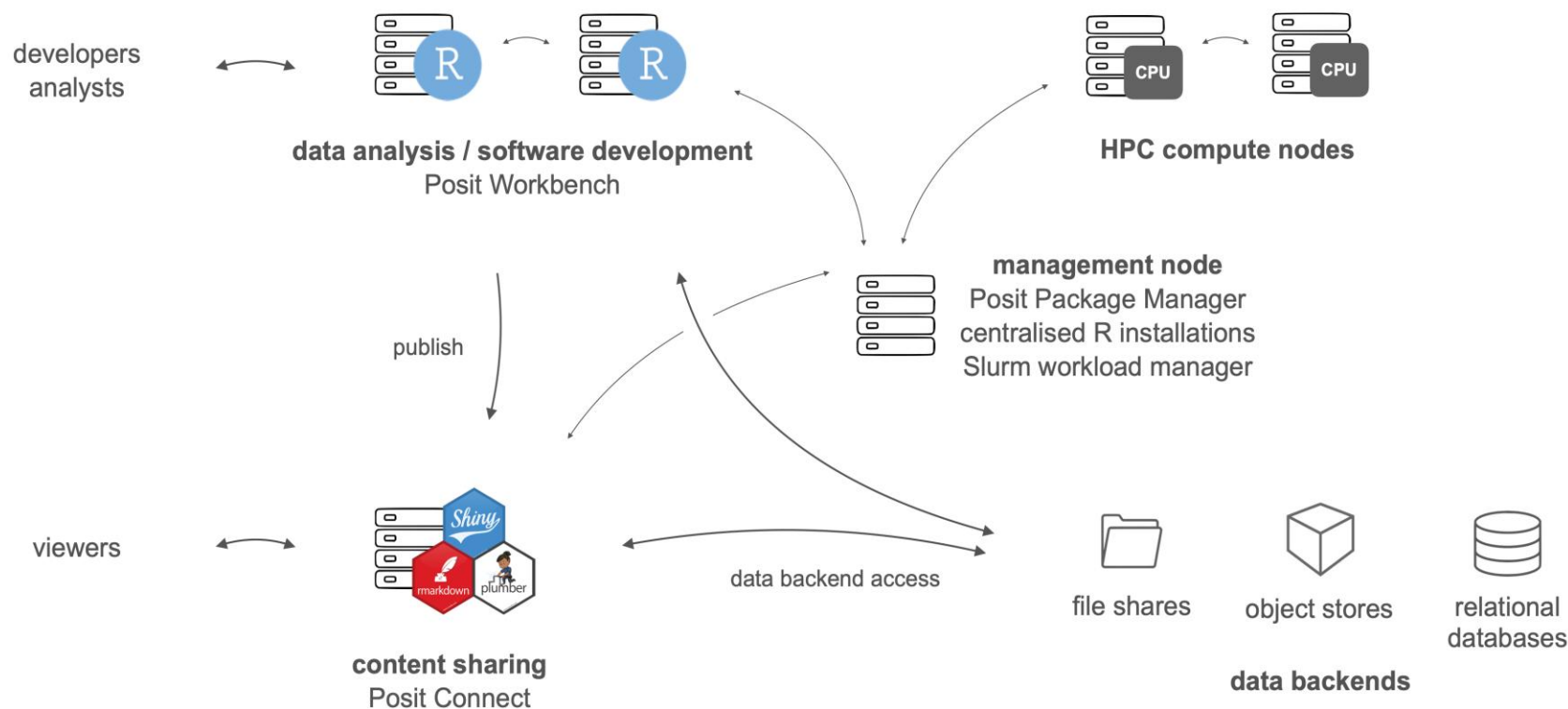
An ecosystem of modular R tools enables scalable, reproducible HEOR modelling across therapy areas

- A continued effort to build an evolving eco-system of R-Shiny based HEOR applications
 - Scope: early-stage economic modelling, based on product profiles
 - Objectives and decisions influenced by early modelling
 - Inputs into multi-country pricing discussions for products early in the pipeline
 - Exploring and communicating the economic value proposition and determining key value determinants
 - Prototyping for future HTA models
 - Early model development options: in-house (internal) or externally built; often as Excel applications
- A decision to build a suite of health economic modelling tools to consolidate and streamline methodological approaches to achieve:
 - Faster iteration and development cycles
 - Reproducible and transparent modelling workflows
 - Reusable, shared tools across therapy areas
 - Integrated into secure enterprise cloud environment
 - Enhanced quality assurance and validation frameworks

Traditional HEOR modelling challenges

Added value	Solutions	Enabling factors
• HEOR modelling has evolved beyond single-purpose tools to complex ecosystems • Organizations require scalable, reusable and consistent approaches across therapy areas and treatments • Efficiency and speed are becoming critical constraints – living model updated. • Increasing expectations for transparency and reproducibility • Potentially - External expectations (HTA, reproducibility)	• Iterative development as the clinical programs and evidence evolve (e.g. Liver disease). Platform model used to guide discussion & further analysis. • Multi-indication and multi-market analysis – builds a database to inform Therapy Area pipeline decision making (e.g. Oncology)	• Acceptance of code-based models (internal, skills driven, & external, regulatory driven) • Rapid learning in code-based model building and collaborative development (largely due to AI tools) • Demonstrable communication capabilities – automated reporting, visual analytical presentations • High performance IT environment for hosting code-based applications

R focused computational platform for biostatistics – for hosting and access to R-shiny HEOR applications



The R focused computational platform in Biostatistics consists of several virtual servers in the cloud and covers three main areas:

- data analysis and software development
- content sharing, eg Shiny web applications and Quarto / R Markdown documents
- High-performance computing (HPC)

Source: User Guide for R-focussed platform

R focused computational platform for biostatistics – for hosting and access to R-shiny HEOR applications



GSK Posit Connect

- Posit Connect allows sharing content like Quarto / R Markdown documents, Shiny web applications, or REST APIs developed with the R package plumber.
- Developers can share content they have generated by publishing it to Posit Connect and granting access to other users.
- Users can view the content by pointing their web browser to the web address assigned to the published content.

For HEOR platforms

- Overall, enabling environment for accommodating HEOR platforms
- Secure GitHub
- High-performance multi-processor computing
- R-Shiny development and publishing
- Database back-end

Source: User Guide for R-focussed platform

Systems in development

Chronic Liver Disease

Several products
Several indications
Multi-country

- Established core liver disease model concept
- Handling complexity of health economic value – clinical evidence and key value drivers
- Suitable to iteration as clinical programme evolves

Oncology Early Modelling Platform

Oncology pipeline
Multiple indications
Multi-country

- Established survival package
- “Standardisation” of partitioned survival model
- Testing the asset profile
- Determining key value drivers and evidence gaps which need investment

Vaccines Value Communication Platform

Several products
Single country
Range of population subgroups

- Existing Excel state-transition models developed into R – automated validation step with millions of model runs.
- User-interface to run models live and display results
- Database to store scenarios unique to each user
- Highly regulated (specific) automated report for dissemination with decision-makers (value communication)

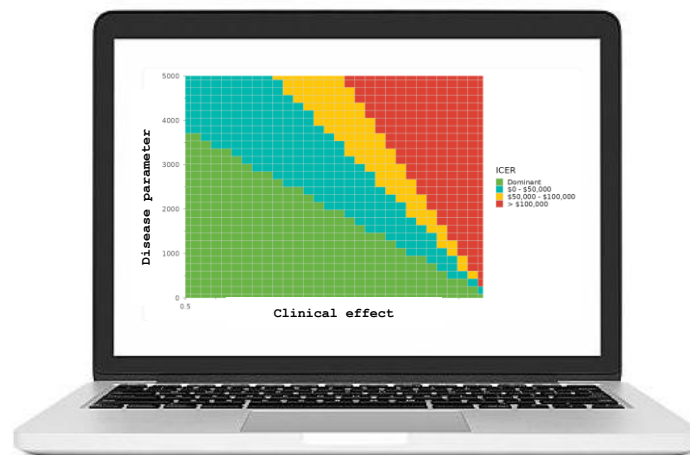
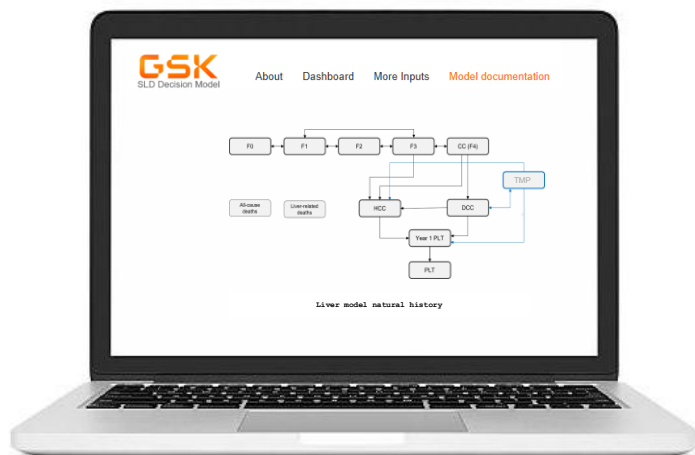
Comm. Disease Microsim

Several products & comparators
Multi-country
Range of population subgroups

- De-novo microsimulation model built, with core functionality validated against existing Excel model
- Model captures individual level treatment & natural history and incorporates behavioural factors.
- User-interface to run models live, display results.
- Database to store scenarios, shared across teams.
- Automated slides, outputs to Excel & .png

Liver Disease Early Economic Platform

Testing product profile and value determinants in a complex chronic disease

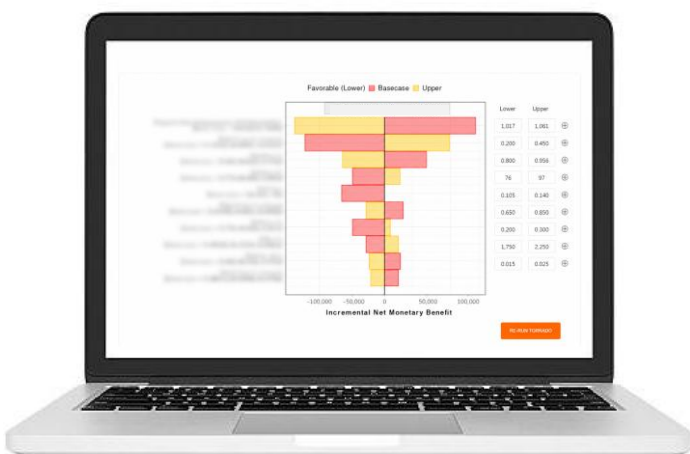


Objectives

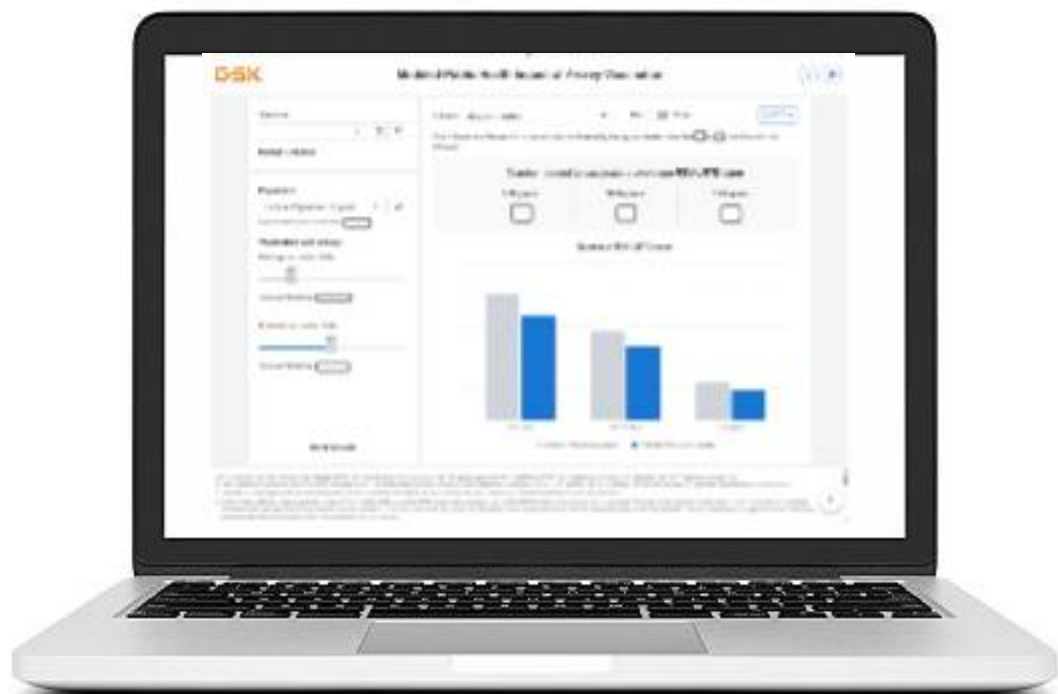
An early-stage economic model aimed to inform the discussions on product profile, economically justifiable price ranges, and key value determinants for several indications and multiple markets.

Features

- Customised GSK branding
- Model parameters read from Excel
- Established state-transition model framework for Liver Disease run in R.
- Sensitivity analysis (DSA/PSA) run in seconds.
- Automated reporting via PDF output
- The platform evolves along the clinical development programme



Value Communication Platform



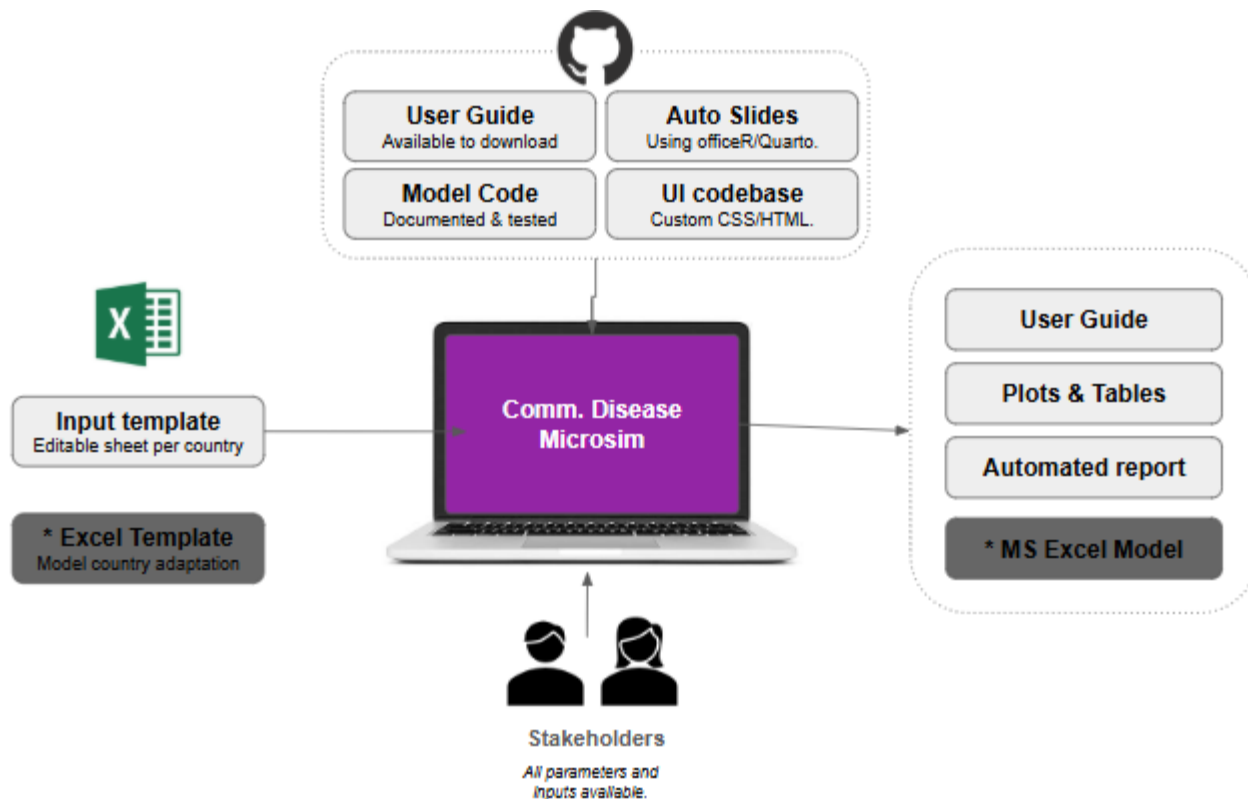
Objectives

The Vaccines Platform is an interactive interface designed to deploy validated epidemiological and HEOR models in a way that makes population-level outcomes explorable beyond the modelling team. The Platform builds on existing, validated model logic recoded from An Excel model into R.

Features

- Scenario management: Save, reload, and compare parameter sets for reproducible analyses.
- Real-time recalculation: Outputs including vaccination numbers, reductions in cases, and deaths, update instantly as inputs change.
- Validation framework: Unit testing and automated Excel–R output comparison.
- Standardised inputs: Structured templates ensure consistent parameterisation.
- User-centered design: Iterative, stakeholder-informed interface for clarity and usability.

Communicable Disease Microsimulation Platform



Objectives

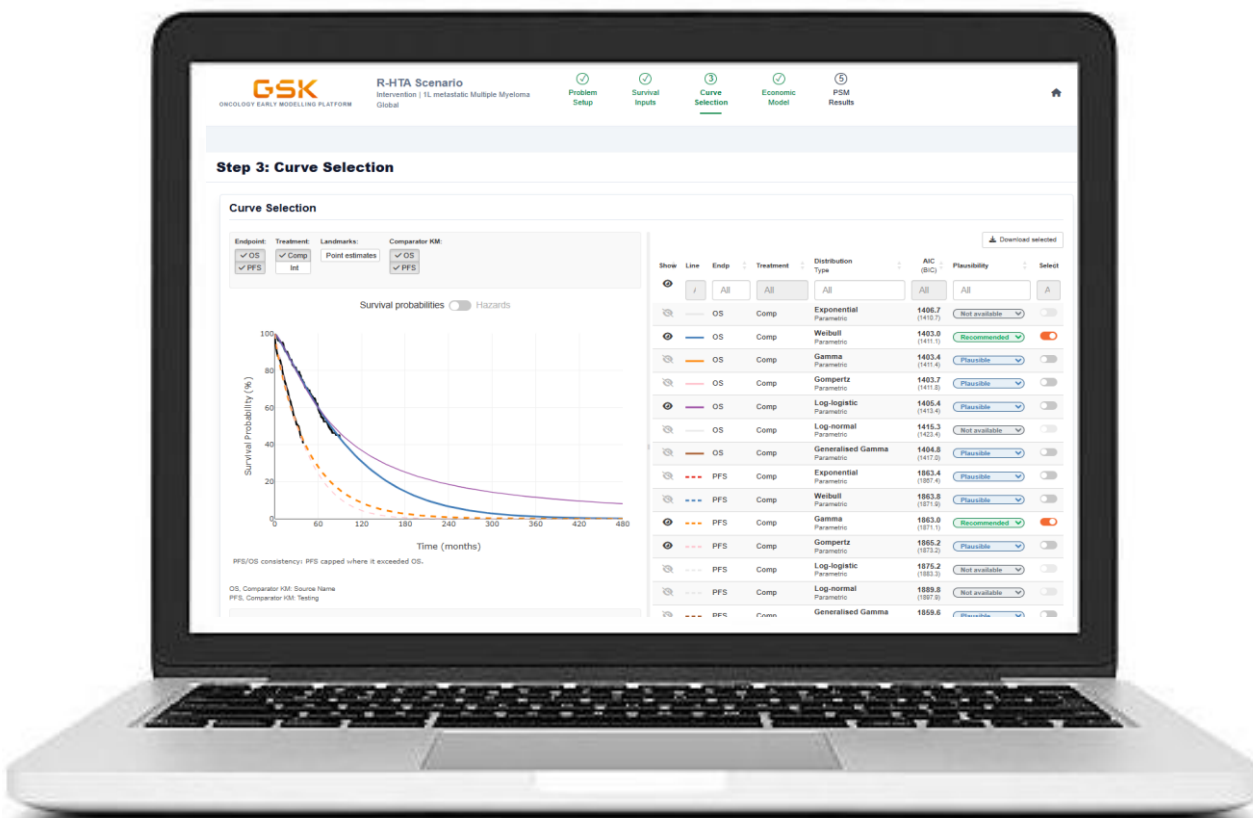
An early-stage economic model designed to meet the aims of the pipeline based on medicine profiles

Enable rapid scenario analysis and evidence generation before full HTA models exist

Features

- High-performance microsimulation model run in seconds.
- Scenarios stored in remote database, including multiple defaults from Excel
- Advanced visualisation and outputs
- Ability to generate downloadable outputs and reports directly from the app
- Scalable and modular platform design

Oncology Early Modelling Platform



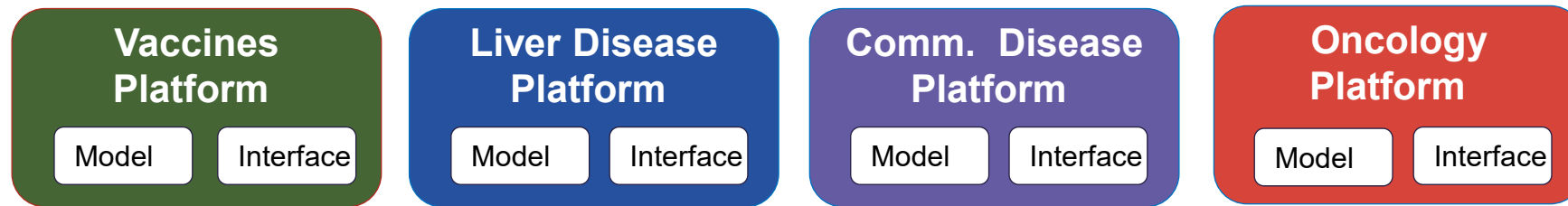
Objectives

An early-stage economic modelling platform designed to meet the aims of the oncology pipeline based on medicine profiles, and published literature

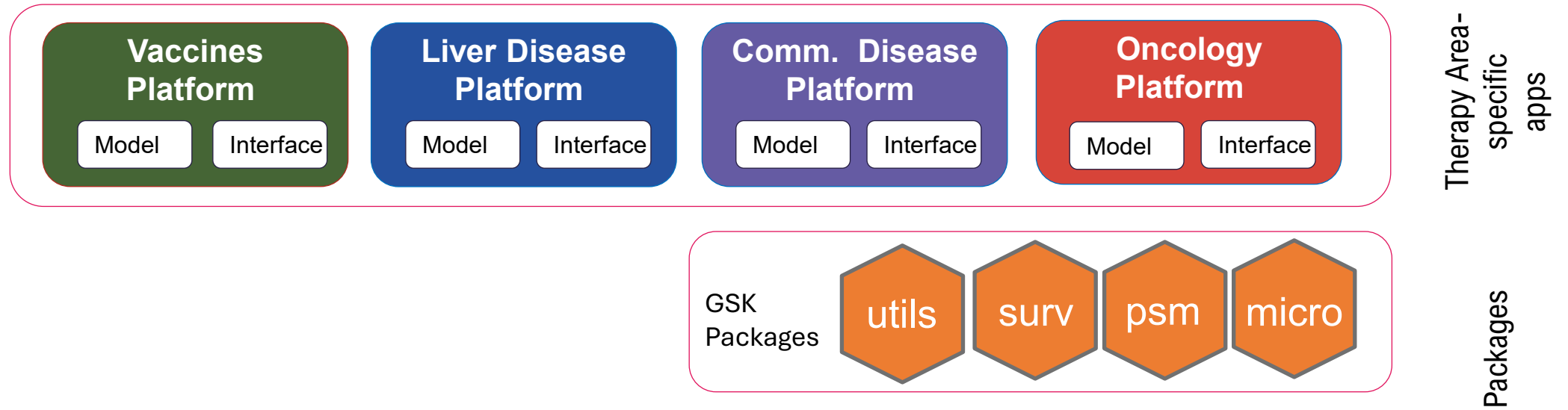
Features

- Partitioned Survival Model framework
- Multi-country analysis across the pipeline – products and indications
- Sequential flow, with multiple validation steps.
- Survival Analysis selection with differing user roles – admin users recommending curves for use in model.
- Automated reporting
- Analyses settings and inputs for each regimen and target population is stored in a database

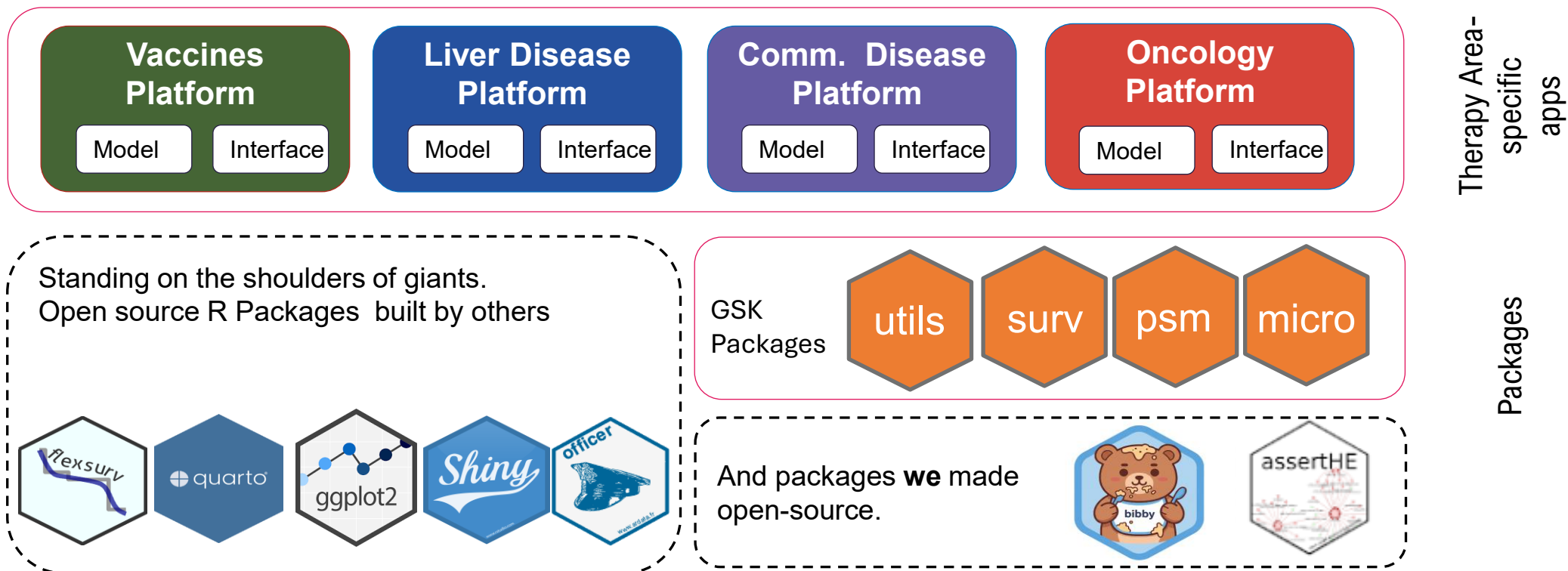
GSK HEOR Modelling Software Ecosystem



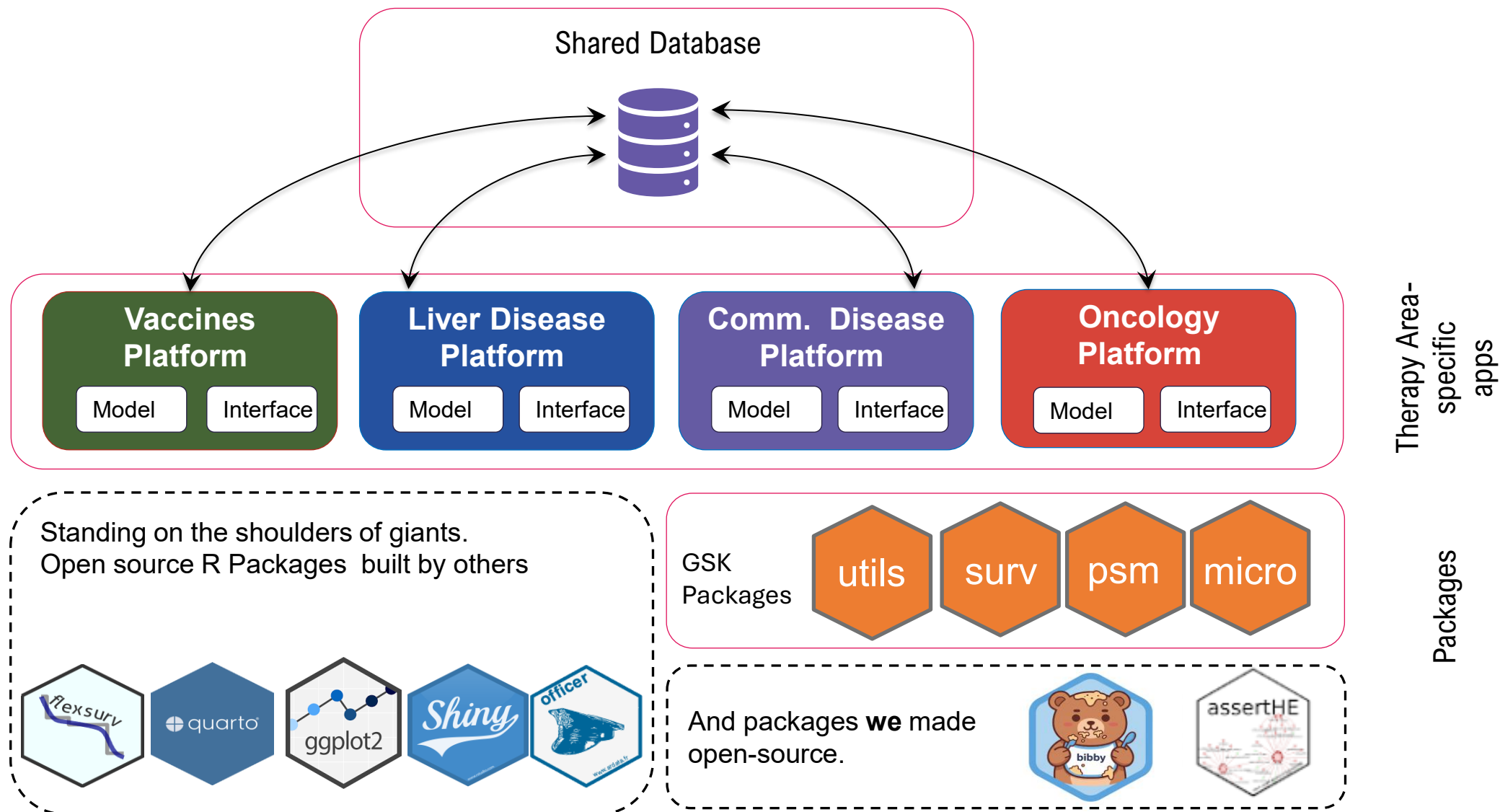
GSK HEOR Modelling Software Ecosystem



GSK HEOR Modelling Software Ecosystem



GSK HEOR Modelling Software Ecosystem



Benefits of this approach – stakeholder perspectives

- Some simulations are not feasible in MS Excel.
- A centralised platform allows for rapid iteration & development
- Look & feel, and outputs, are consistent between projects.
- Analytical team memory developed via database
- Speed! Rapid generation of value insights to inform strategy



Discussion - Summary of learning



Developer's perspective

- Aligning with large company infrastructure requirements
- Learning alongside the organisation, through close, flexible collaboration
- Embracing open-source solutions while carefully managing dependencies
- Piloting prototypes early, recognising issues/bugs and iterating quickly.
- What is the optimal model to deliver HEOR modelling services in the GenAI era - joint development vs software-as service?

Stakeholder/Company Perspective

- Ensuring efficiency and reproducibility of economic models
- Embedding calibration and validation within the model framework
- Facilitating structured expert elicitation and consensus building
- Enabling meaningful customer engagement
- Transitioning from individual product profile models to a reusable knowledge base to inform broader decisions
- Supporting continuous model development alongside the clinical programme
- preparing for potential code-based (R) submissions?